Music Piece Classification

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Revised Data Perturbation

- After incorporating tempo/key perturbations, from results f searching MLP architectures and KNN, accuracies were unreliably high (top-1 accuracies of 0.8775 and 0.8085 respectively)
- The cause was data leakage as perturbed versions of a snippet were in both train/test
  - ex. 0.95 speed version in train, 1.0 speed version in test mean that the model had seen nearly the same thing in train already

Revised Data Perturbation

The method for perturbing was revised from

- **Previous:** perturb full pieces → snippet → assign snippets to train/test
- **Now:** snippet → perturb snippets → assign to train/test all perturbations of a snippet

- Tempo perturbations now [0.8, 0.9, 1.0, 1.1, 1.2] vs [0.9, 0.95, 1.0, 1.1, 1.2] to make more different
- Changed 8s with 4s overlap → 2s overlap to further decrease the data leakage (i.e. less data, may consider making more)
- Note embeddings will be created from variable length snippets

Experiment Pipeline: Motivation

- Prior experiments relied on exploratory notebooks — hard to scale or reproduce
- Refactored into a modular Python pipeline compatible with Prime Intellect reservable instances
- **Core goal:** a model-agnostic pipeline supporting multiple architectures and configs for easy deployment and experimental design

Experiment Pipeline

Designed a modular structure managed with uv

- **run.py**: Entry point — loads configs, orchestrates training & evaluation
- **data.py**: Dataset loading & preprocessing, model-independent
- **train.py**: Training logic with flexible hyperparameter support
- **evaluate.py**: Metrics including Top-k accuracy
- **models/**: Model definitions (MLP baseline + extensible)
- **configs/**: Reproducible experiment configuration

Experiment Pipeline: Benefits of the Refactor

- Rapid iteration on new architectures
- Reproducible experiments across compute environments
- Clean separation of model logic and experiment config
- Scalable execution on Prime Intellect instances
- Foundation for future representation learning and retrieval experiments

Embedding Retrieval: Motivation & Method

Why retrieval?

- Evaluates representation quality rather than just classification accuracy
- Similar musical pieces should cluster together in embedding space

Method

- Generated embeddings per musical segment
- Evaluated via KNN retrieval
- Metrics: Top-1 and Top-5 Accuracy

Results: Initial Dataset

KNN on embeddings

Top-1 Acc: 0.2035

Top-5 Acc: 0.3378

MLP classifier

Top-1 Acc: 0.3440

Top-5 Acc: 0.5640

- KNN significantly underperforms the MLP baseline
- Embeddings contain some structure, but not yet fully discriminative

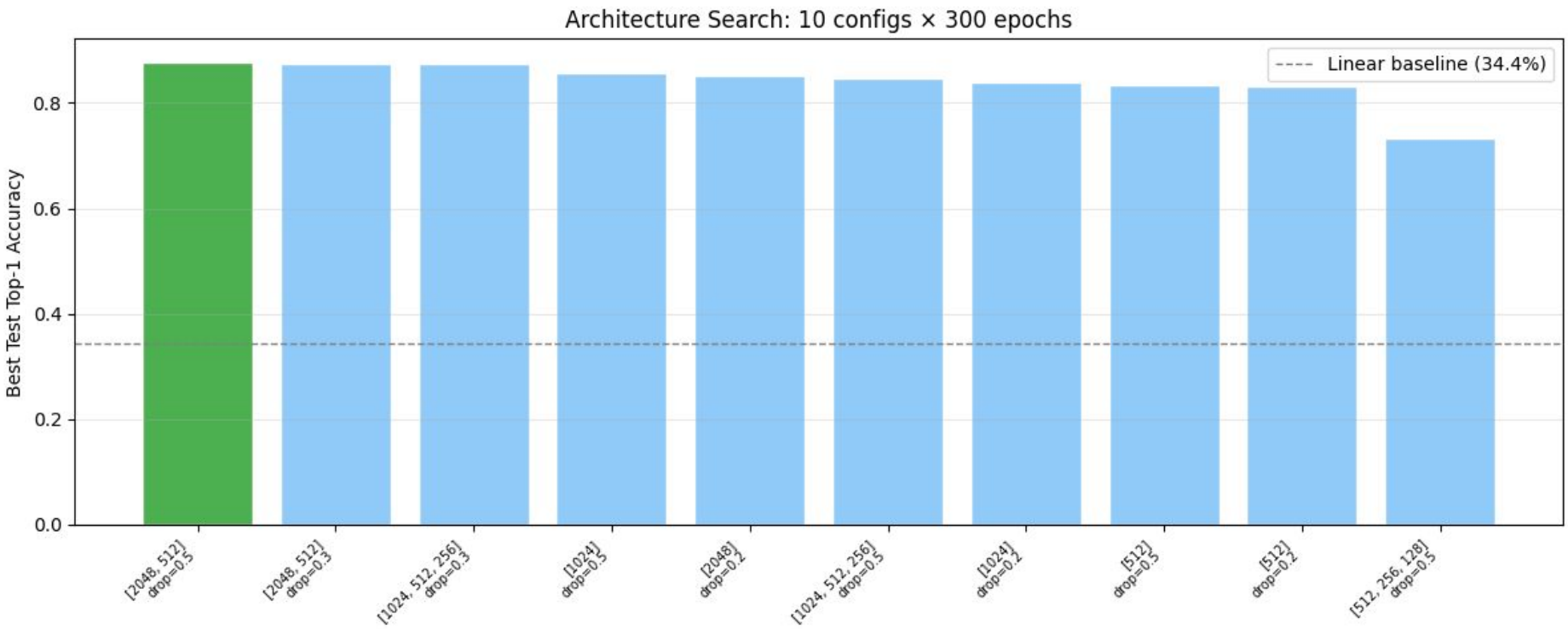
Results: Perturbed Dataset

KNN on embeddings
Top-1 Acc: 0.8085
Top-5 Acc: 0.9325

MLP classifier
Top-1 Acc: 0.8775
Top-5 Acc: 0.9655

- Substantial improvement for both methods
- Gap between KNN and MLP narrowed considerably
- Suggests embeddings meaningfully cluster variations of the same piece

MLP Search Results: Perturbed Dataset



Data Leakage Concern

Caveats with these results

- Perturbed versions of the same piece (e.g. –1 semitone pitch shift) appear in both train and test sets
- These variants are acoustically similar, so the model may be "recognizing" near-duplicate audio rather than learning generalizable representations
- This likely inflates reported accuracy and makes it hard to assess true generalization

Good news: real world analogy

- This setup actually mirrors the task of cover song identification
- In practice, you may want a model that recognizes a piece despite pitch or tempo changes, which is exactly what the perturbation tests

Next step: evaluate with a piece-level train/test split to disentangle leakage from genuine performance

Further Work

1. **Increase Dataset Diversity** - more compositions, styles, and instrumentation
2. **Metric Learning** - triplet loss, contrastive learning, Siamese networks
3. **Alternative Retrieval** - cosine similarity, ANN search, clustering-based methods
4. **Expanded Architectures** - CNNs, transformers, pretrained music models
5. **Scalable Experimentation** - hyperparameter sweeps on Prime Intellect
6. **Extended Perturbations** - broader pitch/tempo range; explore direct embedding perturbations